

LOCAL TIDES

The shape of a coastline can make a big difference to the tidal rise and fall. In some places, such as here in Cornwall, England, it acts like a funnel, concentrating the tidal streams so they cause a very big tidal range—rising very high and falling very low. But other shores have very small tidal ranges, and some have just one tide a day.



RACES AND WHIRLPOOLS

On some coasts, the ebb and flow of the tides can create dangerously fast local currents. This usually happens when the tidal stream has to force its way through a tight gap or around a prominent headland, creating the steep, dangerous waves of a tidal race. Sometimes the flow can swirl back on itself to cause a whirlpool.



◀ SWIRLING WATER

Whirlpools form when opposing currents meet. Large, powerful whirlpools are known as maelstroms.

TIDAL RIVERS

Rising tides also force water up river estuaries. This stops the river flow and may even reverse it. As the river water stops moving, suspended particles settle on the riverbed to form the mudflats that are exposed along many river estuaries at low tide. In some places, a very powerful rising tide can drive a wave of water upriver in a tidal bore. This can be high enough to surf on, and may reach speeds of up to 15 mph (25 km/h).

▶ GLEAMING MUD

As the tide level falls in this river estuary, water draining away down a winding creek exposes a tidal mudflat.

